

Sarus Operation Planner 1.1.0

The ground control station for ArduPilot aircraft from Sarus Aerospace, based on ArduPilot Mission Planner 1.3.83 with all of its screens, features and keyboard shortcuts. This release adds a password lock on the aircraft's setup, takes your parameter values exactly as you set them, and watches both the airframe's limits and the mission in flight. It comes with the second Sarus firmware release, Sarus-2, on ArduPilot 4.7.1 and on 4.6.3.

VERSION

1.1.0

BASED ON

ArduPilot Mission Planner 1.3.83

SARUS FIRMWARE

Sarus-2 on ArduPilot 4.7.1 and 4.6.3

TESTED WITH

Plane / QuadPlane (VTOL), Copter and Rover, in the simulator

WINDOWS

10 or 11, 64-bit

NEEDS

.NET Framework 4.7.2 or later, about 500 MB disk, 4 GB RAM

LICENCE

GNU GPL v3, no warranty

SOURCE CODE

github.com/LikhitS/SarusOperationPlanner

PUBLISHER

Sarus Aerospace Pvt Ltd · sarusaerospace.in

What's new in 1.1.0

The parameter lock

Anyone can still connect, read the parameters, plan and fly. Changing the aircraft's setup now needs the owner's admin password: parameter writes from any page, parameter files, resets, calibrations, firmware install, file writes over

MAVFTP and reboot into the bootloader. Sarus asks for the password once, and changes stay unlocked until you press Lock parameters or close the program.

Aircraft running Sarus-2 firmware enforce the same rule themselves, so another ground station or an older copy of Sarus cannot get round it. After the password, Sarus unlocks each connected aircraft with a signed one-time code. The aircraft holds only public keys, and the password itself is stored nowhere, not even in Sarus. An unlock belongs to one station on one link. It can only be made on the ground but stays valid in flight, and it ends on reboot, on Lock, or after 10 seconds of silence once disarmed.

Only your values

The Full Parameter List no longer shows or flags ArduPilot's recommended ranges, and its spin box no longer pulls a value back into them. Load Presaved and Initial Tune Parameter, which wrote values the program proposed, are gone. After every write Sarus still reads each value back and lists any the aircraft did not keep. We wrote all 3,000 or so parameters of QuadPlane, Copter and Rover and found one real limit in the firmware: loiter acceleration is capped at what the lean angle physically allows, $g \cdot \tan(\text{angle})$. It stays, and Sarus shows it when it applies.

Airframe limits

A new page, CONFIG → Airframe Limits, records what each airframe can do: stall speed, maximum airspeed, climb and descent rates, bank and pitch, the multirotor lean angle and speeds, and for a rover its speed and turn radius. Any parameter that asks for more blinks red in the parameter list as you type it, and in a banner across the window while the aircraft holds it. The value is never changed. The limits are stored per flight controller and understand the parameter names and units of both ArduPilot 4.6 and 4.7.

In-flight alarm

During a mission Sarus compares the altitude, airspeed and track the aircraft is trying to hold with what it achieves. When one stays off by more than the set amount for longer than the set time, and is not closing, a siren sounds and you choose Continue plan, Hold position or Return home. With no answer in 15 seconds the mission continues as planned. Take-off, landing and return legs are not judged.

Look

The badge at the top right now reads "Powered by Sarus Aerospace" beside the logo. Sarus Glass joins the theme list as an option, with a frosted menu bar, a slate window frame, card-style group boxes and softly raised buttons. The default is still Sarus Dark, and the HUD and map keep their colours in every theme.

Sarus firmware Sarus-2

LINE	BASED ON	REPORTS ITSELF AS
4.7	ArduPilot Plane, Copter and Rover 4.7.1	ArduPlane V4.7.1 Sarus-2
4.6	ArduPilot Plane, Copter and Rover 4.6.3	ArduPlane V4.6.3 Sarus-2

On CubeOrangePlus, the 4.6 line offers Plane (without Torquedo boat motors and AIS ship receivers, which a QuadPlane never uses) and Rover; Copter for that board comes from the 4.7 line. Both lines carry the parameter lock and room for the owner's parameter set as firmware defaults, so a new board, a parameter reset or a firmware install starts from the owner's values. Install Firmware lists both lines and asks which one to load.

Reviewed before release

Before release, five separate reviews went through the firmware lock, the program's lock and alarm code, the screens, the changes made in 1.0.0, and the tests themselves. Everything they found is fixed in this release. The password prompt no longer remembers what you type; a Mission Planner input box used to save every entry, passwords included, to the settings file. Write Fast no longer stalls on missions longer than ten items. The alarm window can only be answered with a click, names the aircraft, and confirms Hold and Return with the aircraft. When an aircraft locks itself after a reboot or a silent link, Sarus unlocks it again before your next change instead of letting the change be refused. The firmware now also refuses log erasing, serial passthrough, device writes, calibrations started as auxiliary functions, and parameter changes from transmitter menus while locked, and its unlock codes are harder to predict on older flight controllers.

How 1.1.0 was tested

After the review fixes, the full regression ran against Sarus-2 on 4.7.1 in the simulator and passed all 16 steps: the parameter editor, QuadPlane, Copter and Rover missions, the limits and alarm flight, a flight over a poor link, every keyboard shortcut with a concurrency stress test, every SETUP and CONFIG page, and a write, read-back and restore of every QuadPlane and Rover parameter. On 4.6.3 the lock, alarm, Copter and Rover runs passed as well.

The lock was tested from both sides. The firmware's own tests check each refusal (parameters, every calibration path, every file-write operation, signing setup), wrong keys, replayed, expired and stolen unlock codes, other stations and links, arming, the silence timeout and reboot: 65 checks on each line, all passed, plus 6 checks that firmware built with the owner's keys refuses the public test key. From Sarus, 53 checks on 4.7.1 and 52 on 4.6.3 confirmed that the program's signatures are accepted by the real firmware and that a write which skips the program is still refused. For the alarm, the simulated engine was cut in cruise. The alarm stayed silent through take-off, transition and climb, sounded 11 to 12 seconds after the cut, let the mission continue when nobody answered, and switched to loiter on Hold.

All flights were in ArduPilot's simulator. Test with your own aircraft, radios and flight controller on the bench before you fly, and keep the lock password and its spare in a safe place: nobody can recover them.

What's different from Mission Planner

Everything below was already in 1.0.0, apart from the parameter changes described above.

Replies that went missing

Mission Planner could lose an answer from the aircraft when two parts of the program were waiting at the same time, for example a background camera query while you read parameters. One would pick up the other's reply and throw it away, and you would see a timeout even though the aircraft had answered. Sarus registers for each reply before sending the question, so nothing is taken from it.

OPERATION	BEFORE	SARUS
Parameter reads	6 of 45 lost	0 of 600 lost
Command acknowledgements (arm, mode)	5 of 15 lost	0 of 1,811 lost
Mission upload and download	4 of 10 failed	0 failed
Commands with coordinates	2 of 45 lost	0 of 887 lost

Smaller fixes

The update check no longer complains when there is no update yet, and plugins built by an older version are not reused after an upgrade. A lost legacy rally-point request is retried with the right message. Silent installation with `msiexec /qn` completes instead of hanging on the USB driver wizard.

Privacy, airspace and coexistence

Usage statistics are off unless you turn them on. Sarus installs beside Mission Planner and keeps its own settings, logs and maps. Mission Planner's Altitude Angel plugin is not included and can never load, so the program places no airspace restrictions on a flight; the aircraft's own geofence and failsafes are untouched. The installer allows the program on private and domain networks only, so nobody on public Wi-Fi can reach it, and the built-in web server accepts flight commands only from the PC it runs on.

Known limitations

This release runs on Windows only. The lock cannot stop someone with the flight controller on a bench from flashing other firmware over USB; a secure bootloader is planned for that. Settings that other tools write into DroneCAN peripherals are not covered by the lock. On links others can reach, use MAVLink signing as well, because without it the aircraft knows a ground station only by its link and system ID. Until the installer is code-signed, Windows may say "Windows protected your PC". Choose More info, then Run anyway.

Sarus Operation Planner is based on Mission Planner by Michael Osborne and the ArduPilot contributors. ArduPilot is a trademark of its owners and does not endorse this product.